PingSheng Kao

Santa Clara, California | +1 734 800 6888 | samk840125@gmail.com | linkedin.com/in/pingshengkao

SUMMARY

Research Engineer and ML Data Scientist with 6+ years of experience building evaluation systems for multimodal, LLM, agentic, and production ML systems. Develops simulated users, human-calibrated autoraters, behavioral metrics, trajectory diagnostics, and scalable pipelines that guide model development.

EXPERIENCE

ML Data Scientist - Apple July 2020 - Present
Santa Clara, California

- Built simulated-user and agent-as-judge systems to evaluate an open-ended multimodal conversational agent over long interactions and surface scenario-specific failures.
- Translated human-factors research into task protocols and metrics for per-turn intent misalignment and false updates to inferred user state; evaluated factual consistency and tool-call behavior at scale.
- Built targeted scenario generators and evaluators for visual-memory and sycophancy weaknesses; systematic failure patterns changed subsequent development direction.
- Scaled LLM- and agent-as-judge evaluation on an existing internal batch platform, debugging throughput and rate limits, versioning evaluators, and reporting performance through dashboards.
- Developed a reference-based LLM evaluator and expert/non-expert human-validation workflow used internally for performance reporting and failure discovery; reached 90% expert alignment on 40 examples.
- Implemented large-scale embedding-index experiments to curate segmentation training data from an internal image corpus and delivered the resulting dataset to a foundation-model team.

RESEARCH CONTRIBUTION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks 2025
Experiment contributor (credited as Sam Kao) - accepted to ECCV - arXiv:2512.24592

- Contributed experiments to ECCV-accepted research on grounded, hypothesis-driven error-slice discovery for instance-level vision tasks.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - Ambarella June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor 2018 - 2020
B.S. in Electronic Engineering - National Taiwan University 2013 - 2017
GPA: 3.9/4.0 (B.S.)

SKILLS

Programming & Data: Python, SQL, C++, JavaScript, AWS Athena
Agent Evaluation: Simulated Users, Agent-as-Judge, LLM-as-Judge, Long-Interaction Analysis, Tool-Call Evaluation
Human Alignment: Human Evaluation, Gold Sets, Expert Calibration, Human-in-the-Loop Validation, Intent Alignment
Methods: Metric Design, Scenario Generation, Embeddings, Clustering, OOD Analysis, Experiment Design